

$$\text{Average } M = 2.11 \times 10^{30} \text{ kg}$$

- 3 (a) from the graph, the amplitude of oscillation, x_0 is 2.4 cm or 0.024 m, maximum KE is 75 mJ. Since mass of trolley is 0.590 kg, then $\text{KE}_{\text{max}} = \frac{1}{2} m(v_{\text{max}})^2$

$$0.075 = \frac{1}{2} (0.590)(\omega x_0)^2$$

$$\omega = 21.0 \text{ rad s}^{-1}$$

$$f = 3.34 \text{ Hz}$$

- (b) loss of energy but the frequency of oscillation is unchanged (hence $\omega = \text{constant}$).
The whole graph shifted down by 40 mJ. The new intersection of the graph with the x-axis will be the new amplitude. $x_0 = 1.65 \text{ cm}$

(The peak of the graph falls from 75 mJ to 35 mJ)

